

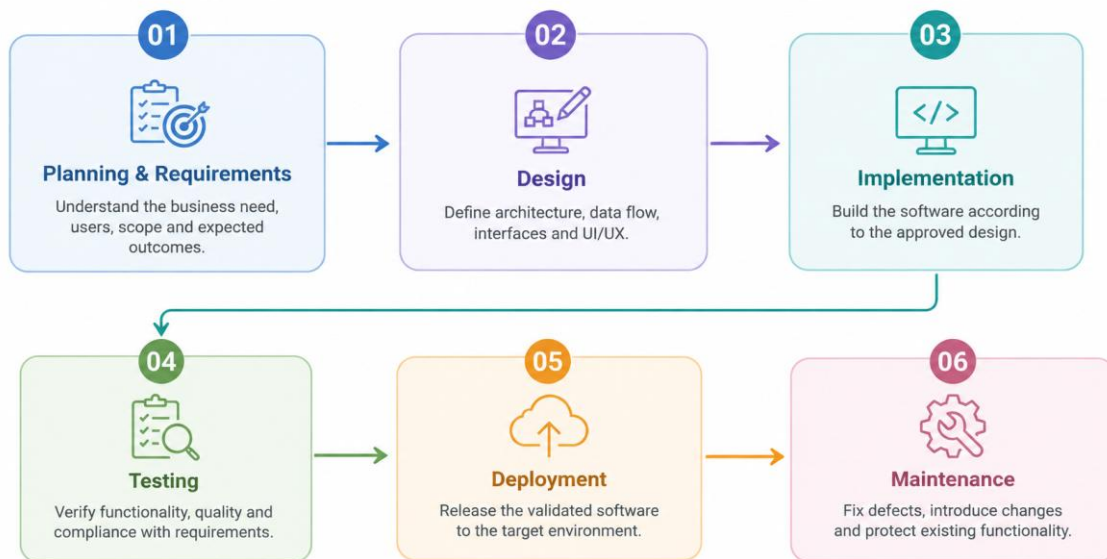
MODULE 2 — ASSIGNMENT

Software Development Life Cycle (SDLC) in the Age of AI

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Course: AI Driven SQA — Manual & Automation Testing | OSTAD

Q1. SDLC Lifecycle Diagram & Workflow Explanation



Workflow explanation:

1. Requirement Analysis — the team talks to stakeholders to find out what the business and users need. These needs are written down as functional and non-functional requirements in a BRD/SRS. AI can help by summarizing meeting notes into a first draft of the requirements.
2. Design — once requirements are approved, the team plans how the system will be built. This includes the High-Level Design (overall system) and Low-Level Design (module details), plus database, API, and UI/UX design. AI can suggest a first draft of the architecture or database schema, which the team then checks and adjusts.
3. Implementation — Developers write the actual code, following the approved design. They also do unit testing and use Git for version control. AI coding tools like GitHub Copilot or Cursor AI can suggest code as it is written, saving time on repetitive parts.
4. Testing — QA checks that the software works the way it should and finds any defects before release. This includes test case design, defect reporting, and regression testing. AI can help generate test cases or sample test data, but a human still needs to check that the important scenarios are covered.
5. Deployment — the tested software is released to the live environment, along with smoke testing and monitoring. AI-based monitoring tools can help spot unusual activity (for example, a sudden spike in failed transactions) faster than checking logs manually.
6. Maintenance — after release, the team fixes bugs, applies security updates, and improves the software based on user feedback. AI can help by pointing out patterns in past issues, similar to how a support team keeps track of common customer complaints.

Q2. Practical AI Exercise

AI tool used: Claude (Anthropic).

2.1. Question: Generate five user stories for an E-commerce application.

Five User Stories: E-Commerce Application

ID	Feature	User Story
US-01	Product Search	As a customer, I want to search products by keyword and category, so that I can quickly find the items I want to buy.
US-02	Shopping Cart	As a customer, I want to add items to a shopping cart, so that I can purchase multiple products in a single checkout.
US-03	Multiple Payment Methods	As a customer, I want to pay using multiple payment methods (card, mobile banking, or cash on delivery), so that I can choose whichever option is most convenient for me.
US-04	Order Tracking	As a customer, I want to track my order status in real time, so that I know when to expect delivery.
US-05	Inventory Management	As an admin, I want to manage product inventory (add, edit, or remove stock), so that customers only see items that are actually available.

2.2. Question: Convert one user story into at least five acceptance criteria.

Acceptance Criteria (Story #1 : Product Search)

Story #1 — Product Search

User Story As a customer, I want to search products by keyword and category, so that I can quickly find the items I want to buy.

ID	Scenario	Acceptance Criterion
AC-01	Search by Keyword	Given a customer is on the product search page, when they enter a valid product keyword, then the system should display products matching the keyword.
AC-02	Search by Category	Given a customer selects a product category, when the search is performed, then the system should display products available in that category.
AC-03	No Matching Product	Given a customer enters a keyword that does not match any product, when the search is completed, then the system should display a clear message that no matching product was found.
AC-04	Relevant Search Results	Given a customer searches for a product, when the results are displayed, then the results should be relevant to the entered keyword or selected category.
AC-05	View Product Details	Given a customer receives search results, when they select a product, then the system should open the product details so the customer can review the product before purchasing.

QA Coverage The criteria cover positive search, category filtering, no-result handling, result relevance and product-detail access.

2.3. Question: Ask the AI to create a Sprint Backlog

AI-Generated Sprint Backlog

Product Search Epic

Epic Focus Build and validate a reliable product search experience that allows customers to find products by keyword and category.

Task ID	Backlog Item	Owner	Estimate (SP)	Priority
PS-01	Define product search requirements	Business Analyst / Product Owner	3	High
PS-02	Design search and filter UI	UI/UX Designer	3	High
PS-03	Develop keyword search functionality	Backend Developer	5	High
PS-04	Implement category-based search	Backend Developer	5	High
PS-05	Develop search results UI	Frontend Developer	5	High
PS-06	Handle no-result search scenarios	Frontend Developer	3	Medium
PS-07	Write & execute search functional test cases	QA Engineer	5	High
PS-08	Automate product search regression tests	QA Automation Engineer	5	Medium
PS-09	Perform end-to-end search testing	QA Engineer	3	High
PS-10	Conduct UAT for product search	QA / Product Owner	3	High

Backlog Review

- High-priority items focus on the core product-search journey and its QA validation.
- Negative scenarios include no-result handling.
- QA work covers functional, regression and end-to-end testing.
- UAT confirms that the completed search flow meets customer and business expectations.
- Story-point estimates are planning estimates and should be reviewed against the team's actual capacity and velocity.

Backlog Alignment User Story → Acceptance Criteria → Sprint Backlog → QA Validation → UAT

2.4. Review of AI-Generated Output & Manual Improvements

What the AI did well

- Generated five user stories using the standard “As a [role], I want [goal], so that [benefit]” structure.
- Converted the selected story into Given/When/Then acceptance criteria.
- Produced a sprint backlog with task IDs, owners, story-point estimates and priorities.
- Created a useful starting structure quickly, reducing the time needed to organize the exercise.

Corrections and improvements made manually

Area	Manual Improvement
Acceptance criteria	Separated keyword and category search into two independently testable scenarios.
Negative coverage	Added a clear no-results criterion and refined the backlog to remove invalid-search handling that was not covered by the acceptance criteria.
Testability	Refined the result-relevance criterion so that each displayed result must satisfy the selected search condition.
Backlog priority	Kept requirements and core QA validation at High priority because the feature depends on clear requirements and end-to-end verification.
Traceability	Aligned the backlog with the selected Product Search story and its acceptance criteria.

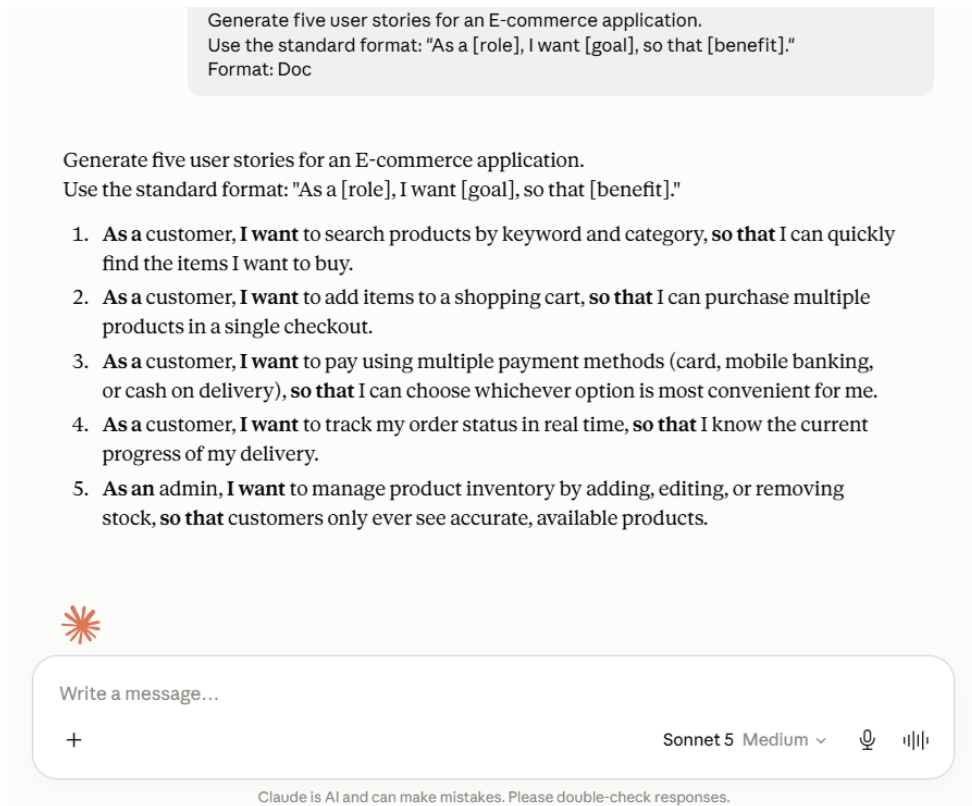
Why human review is still necessary

- AI does not know the actual product catalog, category rules, search logic or business priorities unless these are provided.
- AI can produce confident-looking output while missing edge cases or using requirements that are too broad.
- Story-point estimates and priorities are planning suggestions and need to be checked against the team's real capacity and velocity.
- Human review is required to confirm correctness, testability, business fit and overall software quality.

2.5. Screenshots of AI Interaction

Screenshot (I) — Claude Interaction: Five E-commerce User Stories

Actual Claude conversation showing the prompt used to generate five e-commerce user stories and the generated response.



Screenshot (II) — Claude Interaction: Acceptance Criteria and Sprint Backlog

Actual Claude conversation showing the prompt used to generate five acceptance criteria and the Sprint Backlog for User Story #1 — Product Search.

Part 1 of 3

Convert User Story #1 — Product Search into 5 clear acceptance criteria, then create a Sprint Backlog based on the same user story and acceptance criteria. Include Task ID, Backlog Item, Owner, Story Point Estimate, and Priority.

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User Story #1 — Product Search

As a customer, I want to search products by keyword and category, so that I can quickly find the items I want to buy.

Acceptance Criteria

ID	Scenario	Acceptance Criterion
AC-01	Search by Keyword	Given the customer is on the product search page, when a valid keyword is entered, then the system should display products that match the keyword.
AC-02	Search by Category	Given the customer selects a category, when the search is performed, then the system should display products belonging to the selected category.

Part 2 of 3

AC-03	No Matching Product	Given no product matches the entered keyword or category, when the search is completed, then the system should display a clear no-results message.
AC-04	Relevant Search Results	Given matching products exist, when the customer searches by keyword or category, then every displayed result should satisfy the selected search condition.
AC-05	View Product Details	Given search results are displayed, when the customer selects a product, then the system should open that product's details page.

Sprint Backlog

Task ID	Backlog Item	Owner	Estimate (SP)	Priority
PS-01	Define product search requirements	Business Analyst / Product Owner	3	High
PS-02	Design search and filter UI	UI/UX Designer	3	High
PS-03	Develop keyword search	Frontend Developer	5	High

Part 3 of 3

PS-04	Implement category-based search	Backend Developer	5	High
PS-05	Develop search results UI	Frontend Developer	5	High
PS-06	Handle no-result and invalid-search scenarios	Frontend Developer	3	Medium
PS-07	Write and execute search functional test cases	QA Engineer	5	High
PS-08	Automate product-search regression tests	QA Automation Engineer	5	Medium
PS-09	Perform end-to-end search testing	QA Engineer	3	High
PS-10	Conduct UAT for product search	QA / Product Owner	3	High

Note: Due to the length of the AI-generated response, Screenshot 2 was captured across three continuous sections to ensure the complete output is included.